Inputs/Additions

1. Medicinal plant identification
Let users upload a leaf photograph or capture one using their mobile camera. The AI should identify the plant and display its common name, scientific name, family, and key identifying characteristics.
Example: Identify Tulsi, Neem, Aloe vera, Mint, and other supported medicinal plants.

2. AI-based leaf disease detection
Identify visible symptoms such as brown spots, yellowing, fungal growth, and leaf damage. Display the predicted condition, confidence score, possible causes, and preventive measures.
The system should distinguish disease detection from general plant identification.

3. Explainable AI – show why the prediction was made
Highlight the part of the leaf that influenced the AI prediction, using a heatmap or visual overlay. Explain which visible features contributed to the result.
This makes the project more technically interesting than simply displaying a plant name. The highlighted area should be presented as an explanation, not proof that the diagnosis is correct.

4. Regional-language AI assistant
Allow users to read or hear the results in Kannada, Hindi, English, and other relevant languages. Include a voice-based question-and-answer facility.
This can make the application more accessible to farmers, home gardeners, and rural users.

5. Personalized plant-care recommendations
Based on the identified plant, provide guidance on watering, sunlight, soil, drainage, seasonal care, and common pests. Ask users for their location or growing conditions when needed.
Recommendations should be specific to the identified plant rather than generic advice for every plant.

6. Plant history and health tracking
Save previous scans, dates, predictions, and photographs. Let users compare images over time to observe changes in plant health.
A history dashboard makes the application useful beyond a one-time demonstration.

7. Weather-based disease-risk alerts
Use local temperature, humidity, rainfall, and suitable disease-risk rules to indicate when a particular plant may be more vulnerable to certain diseases.
Present this as a risk estimate, not as a confirmed diagnosis. Start with a small number of well-supported plant-disease combinations.

Unknown plant detection
Recognize when an image is outside the model's supported classes and avoid giving a misleading identification.

Image quality checker
Medicinal plant knowledge base

Offline or low-network mode
Explore a lightweight model that can run on a mobile device without a constant internet connection.

Expert feedback facility
Let users report incorrect predictions and, where possible, submit verified corrections for future model improvement.

Important safety point:
Since this project includes medicinal plants, it should not claim that identifying a plant makes it safe to consume or use as a treatment. Some plants look alike, and incorrect identification can be dangerous. The application should clearly communicate uncertainty and avoid prescribing medicinal dosages.